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Nicholas Reardon

Gameplay and Systems Programmer

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Summary

Gameplay and systems programmer building production-style Unreal Engine 5 systems on Lyra, focused on C++, modular data-driven architecture, combat, encounters, persistence, and designer-facing workflows.

Skills

Languages: C++, C#, Python, GDScript, Java, SQL

Engines and Frameworks: Unreal Engine 5, Unity, Godot, Qt

Unreal Engine: Gameplay Ability System, Game Feature Plugins, Lyra, Enhanced Input, Behavior Trees, EQS, UMG, CommonUI

Gameplay and Systems: Data-driven Design, Modular Combat, Encounter Systems, Spawn Systems, Save and Persistence, AI State Machines, Authoring Tools

Tooling: Git, Diversion, JetBrains Rider, Visual Studio, UnrealVS

Projects

Tethered - Case study — Sept 2025 – present

- Architected a modular Lyra-based gameplay framework using Game Feature plugin isolation to keep combat, game modes, and content decoupled while iterating on an arena-mode vertical slice.
- Designed a world-level event bus on the GameState Ability System Component to coordinate encounter, spawn, and travel systems at runtime without coupling individual components.
- Built a seamless-travel subsystem that snapshots and reconstructs player and world state across level transitions, preserving GAS attributes, equipment, and run upgrades through GameState teardown.
- Implemented a pluggable save backend behind an interface, isolating disk I/O from runtime state and leaving a clean extension point for a future server-backed implementation.
- Designed a designer-facing combo system using an input-tag routing map where each step resolves attacks from owned Gameplay Tags at runtime, letting upgrades modify movesets without branching logic.

Alien Survivors - Playable — Sept 2024 – Dec 2024

- Built an enemy spawning director with spawn pacing, cap management, and optional object pooling for performance under high enemy counts.
- Implemented rarity-weighted enemy selection with per-type parameters and runtime tuning hooks for spawn rate and movement speed.

Last Oasis - Playable — 2025 game jam

- Built a nested state-machine character controller with ground, air, wall-slide, slam, destroy, and death states; coyote time; variable-height jumping; wall jumps; input buffering; spike grace frames; and tile-data-driven slippery surfaces.
- Built a chain-reaction destructible system where a facing raycast breaks a target and an expanding overlap query breaks neighbors outward with per-depth delay.
- Created an editor-time tile-to-scene tool that swaps tiles for scene instances keyed by tile custom data while preserving flip, transpose, and rotation values.
- Built an aimable launch ability with aim, fire, and cancel states, a live guide line, and on-screen control prompts.

A Totally Normal Bike Ride - Playable — 3-day game jam

- Built a two-body physics character using a wheel and rider rigidbody joined by a pin joint, with torque-based locomotion and separate ground and air balance tuning.
- Implemented a charge-and-release variable jump with compression, impulse scaling, contact-based ground state, and rolling average-speed feedback.

Education

California State University, Fullerton, Bachelor of Science in Computer Science – Fullerton, CA — Dec 2026

- Expected graduation: Fall 2026.

Saddleback College, Associate of Science in Computer Science – Mission Viejo, CA — May 2024

- Graduated May 2024.

Experience

Electronics Support Technician, Cheapest iPhone Repair – Laguna Niguel and Irvine, CA — June 2023 – July 2024

- Independently diagnosed and repaired unfamiliar devices through structured troubleshooting, fault isolation, and verification testing.
- Executed delicate device teardowns with detailed tracking and documentation for reliable reassembly.
- Built internal repair documentation and improved workflows to increase consistency and efficiency.